1. Introduction

With the rapid development and popularization of drone technology, drones have shown great application potential in aerial photography, agriculture, rescue, security and other fields, but at the same time they have also brought increasingly serious safety risks. Incidents such as illegal intrusions, malicious reconnaissance, and privacy violations occur frequently, posing serious threats to national security, public safety, and personal privacy. Therefore, the development of efficient and advanced drone jamming technology has become an urgent need to maintain airspace security and ensure social stability.

2. Device Overview

The fixed DR100-A drone detection and jamming device consists of a drone detection system and a radio interference system, which can detect, identify, interfere with drones. It can make the drone hover, make an emergency landing or return, effectively protecting the safety of low-altitude airspace and preventing the drone from entering the defense zone. It has excellent detection and defense performance, long-distance working ability, no need for personnel on duty.

2.1 Device Name

Fixed DR100-A drone detection and jamming device

2.2 Device Model

DR100-A

2.3 Device Usage

Incidents such as illegal intrusions, malicious reconnaissance, and privacy violations occur frequently through drones, posing a serious threat to national security, public safety, and personal privacy. Therefore, the development of efficient and portable drone countermeasures technology has become an urgent need to maintain airspace security and ensure social stability.

2.4 Device Technical Parameters

No.	Name	Performance indicators
1	Detection frequency bands	70MHz-6GHz full frequency band scanning, detection and display
2	Key detection frequency bands	800MHz, 900MHz, 1.2GHz, 1.4GHz, 2.4GHz, 5.2GHz, 5.8GHz
3	FPV analog video transmission signal detection	500MHz~6GHz scan and display
4	FPV analog video transmission signal detection distance	≥1.5km (support viewing real-time analog video)
5	Detection distance	≥10km (depends on working conditions)
6	Detection range	The horizontal detection angle can reach 360° Vertical detection range: -90° ~ 90°
7	Detection response time	The time required from when the equipment is powered on to when it discovers and reports the drone ≤ 10s
8	Direction finding accuracy	≤±3° (RMS, Under site conditions with no obvious obstructions or obvious electromagnetic interference)
9	Detection display	Drone distance, flight altitude, speed, position (including longitude and latitude, and required to be accurate to 1 meter), drone model, drone SN code, drone take-off point, drone return point, drone real-time trajectory, etc.

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No.	Name	Performance indicators
10	Jamming distance	≥3km (depends on working conditions)
11	Interference azimuth coverage	The horizontal detection angle can reach 360°
12	Jamming frequency bands	Omnidirectional jamming frequency bands: 420MHz-480MHz 50W 650MHz-730MHz 50W 740MHz-840MHz 50W 860MHz-930MHz 50W 1080MHz-1200MHz 50W 1200MHz-1340MHz 50W 1340MHz-1500MHz 50W 1560MHz-1620MHz 50W 4880MHz-5080MHz 50W 5100MHz-5300MHz 50W 5870MHz-6060MHz 50W 5720MHz-5860MHz 100W 2385MHz-2490MHz 100W
13		Directional jamming frequency bands: 850MHz-950MHz 50W 2385MHz-2490MHz 50W 5720MHz -5860MHz 50W 5100MHz-5300MHz 30W
14	Jamming response time	≤7s
15	Cloud platform	(1) Unattended and manual control functions: The system can automatically complete drone detection,

No.	Name	Performance indicators
		identification, alerting, localization, and mitigation. The software automatically activates jamming devices, and based on the drone's location information from monitoring devices, it provides fully automated tracking interference. (2) Blacklist and whitelist functions (3) High-definition GIS system: It presents surrounding environment information, device deployment points, and other relevant details. (4) Equipped with log playback and statistical records: drone intrusion log, the log displays angle, alarm time, frequency information, drone intrusion records can be played back, and drone intrusion statistical reports can be provided; (5) Positioning and automatic north calibration: The system features GPS positioning, allowing the location to be marked on a map, and it is capable of automatic north calibration. (6) It has drone signal extraction and addition functions: it can automatically collect unknown drone signals, automatically extract parameters, and add them to the drone feature library through manual confirmation; (7) It has the ability to display device offline, standby, power-off, and abnormal status.
16rr	Jamming device (omnidirectional) *2	Weight: single device≤31kg
17		Size: Length 520mm*width 465mm*height 275mm
18		Description:

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No.	Name	Performance indicators
		Main device and secondary device (see the logo) to achieve omnidirectional jamming
19	Jamming device (directional)*4	Weight: single device≤5kg
20		Size: Length320mm*width 320mm*height 110mm
21		Description: Divided into four sides, four host devices in total (see the logo), to realize directional jamming in key frequency bands
22	Detection device weight	≤15kg
23	Detection device size	Diameter 400mm*height 370mm
24	Device total weight	≤150kg
25	Device total size	Diameter 1000mm*height 2400mm
26	Working temperature	-40℃~60℃

3. Device Features and Functions

The fixed DR100-A drone detection and jamming device consists of a drone detection system and a radio interference system, which can detect, identify, interfere with drones. It can make the drone hover, make an emergency landing or return, effectively protecting the safety of low-altitude airspace and preventing the drone from entering the defense zone. It has excellent detection and defense performance, long-distance working ability, no need for personnel on duty.

Full frequency band detection: 70MHz-6GHz full frequency band detection, covering all drone models.

Super interference with FPV drones: Effective defense against radio frequency flight control and FPV drones, with outstanding interference effect.

Simulated image transmission interference capability: Effectively prevents the intrusion of drones using simulated image transmission

Multiple interference modes: Directional interference in key frequency bands and omnidirectional interference covering all frequency bands, effectively preventing the intrusion of high-power drones and non-standard frequency band drones.

Strong scalability: Expand and integrate radar, optoelectronics, navigation and deception and other equipment, support work by one equipment and by multiple equipment network .

Unattended: Realize all-weather automatic and intelligent execution of preset plans. When a drone is detected, it can quickly coordinate strikes outside the defense zone.

Military quality: Covered by aviation aluminum fuselage, no fear of high and low temperatures, strong environmental adaptability.

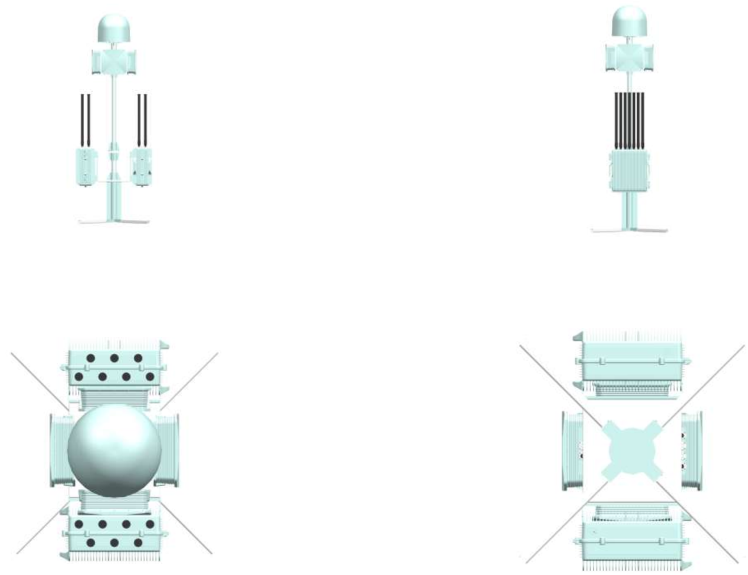
Signal collection: Built-in drone signal collection function, which can collect and learn unknown signals.

Customized services: Customized intelligent solutions according to needs to provide customers with better defense effects.

4. Device Structure and Components

The fixed DR100-A drone detection and jamming equipment consists of a drone detection system, a radio interference system, a control motherboard, a support rod assembly, main body shell, and power supply assembly, etc.

4.1 Device Four Views



4.2 Device List

No.	Name	Picture	Number	Size (mm)	Weight (kg)
1	Detection host		1	Diameter 400*height 370	13.5
2	Jamming host		2	Length 520*width 465*height 275	30.25
3	Directional jamming host		4	320mm*320mm*110mm	4.85
4	Jamming antenna		13	Diameter 32*height 600	0.2
5	Support rod assembly		1	Diameter 50*height 500	46

6	Cable		12	/	0.15
7	Detection host aviation case		1	680mm*600mm*480mm	15.05
8	Omnidirectional jamming host aviation case		2	470mm*570mm*710mm	27.5
9	Support rod assembly aviation case		1	1700mm*440mm*350mm	12.5
10	Directional jamming host aviation case		1	620mm*430mm*430mm	7.9

5. Device Control and Interface

Product interface definition:

Detection device power supply interface: 3-core power cord (AC220V 50Hz)

Detection device data interface: LAN (RJ45), WAN (RJ45), DATA (9-core)

Directional jamming device power supply interface: 3-core power cord (DC28V 50Hz)

Directional jamming device data interface: DATA (9-core)

Omnidirectional jamming device power supply interface: 3-core power cord (AC220V 50Hz)

Omnidirectional jamming device data interface: DATA (9-core)

5.1 System Operation Interface

5.2 Device Security Protection

Aviation Case：

Omnidirectional host packaging size: 470mm*570mm*710mm

Directional host packaging size: 620mm*430mm*430mm

Detection packaging size: 680mm*600mm*480mm

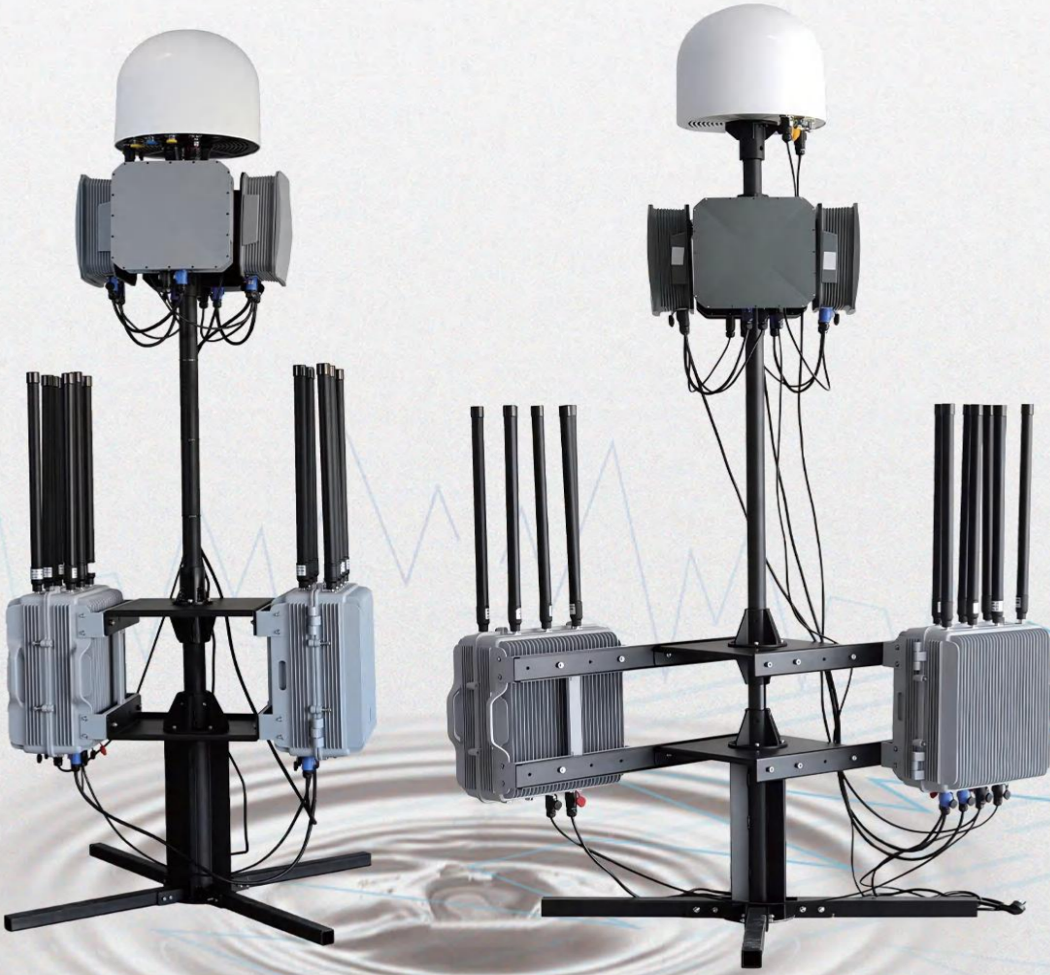


6. Device Maintenance

Power on the device regularly to check its status.

DR100-A

Fixed Drone Detection and Jamming Device



13 Frequency Bands & 4 Directional Jamming Arrays
Super Interference to FPV Drones | 1+N Flexible Deployment
FPV Video Transmission Detection and Capture
360° Full Frequency Band Detection

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Introduction

The device has excellent detection, identification, interference, jamming and long-distance working capabilities. It does not require supervision and automatically detects and counteracts drones, letting them to hover, land or return. It prevents drones from entering the defense zone and protects the safety of low-altitude airspace.

Features

- Full Frequency Band Detection
- FPV Video Transmission Jamming
- Multiple Jamming Modes
- 1+N Equipment Interlinking
- Super Interference to FPV Drones
- FPV Video Transmission Signal Capture

Specification

Detection frequency band	70MHz-6GHz full-frequency scanning, detection and display
Detection range	≥10km(depends on working conditions)
FPV Video transmission detection	500MHz-6GHz full-frequency scanning and display
Video transmission detection range	≥1.5km(supports viewing real-time video)
Azimuth range	Horizontal: 360° Vertical:-90°-90°
Detection response time	≤10s
Omnidirectional jamming frequency band	420MHz-480MHz, 650MHz-730MHz, 740MHz-840MHz, 860MHz-930MHz, 1080MHz-1200MHz, 1200MHz-1340MHz, 1340MHz-1500MHz, 1560MHz-1620MHz, 4880MHz-5080MHz, 5870MHz-6060MHz , 5100MHz-5300MHz, 5720MHz-5860MHz, 2385MHz-2490MHz(customizable)
Directional jamming frequency band	850MHz-950MHz, 2385MHz-2490MHz, 5720MHz -5860MHz, 5100MHz-5300MHz
Jamming distance	≥3km (depends on working conditions)
Operation temperature	(-40°C~+60°C)±2°C
Weight	≤150kg
Size	1000mm*2400mm (diameter*height)